# Sophie Zhao

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## EDUCATION

**Carnegie Mellon University,** Pittsburgh, PA — Expected May 2030
Bachelor of Science in Computational Physics, Intended Additional Major in Computer Science
Relevant Courses: Matter and Interactions (computational mechanics), Concepts of Mathematics & Proofs (discrete math)

**Stuyvesant High School,** New York, NY — June 2026
Stuyvesant Endorsed Diploma
Overall GPA: 98/100

## ACTIVITIES

**Co-President - High School Astronomy Club** — September 2023 - June 2026
- Raised $1500 to fund club telescope, binoculars, and trips to observatories and museums
- Collaborated with NASA ASTROCAMP and expanded club membership from 20 to 110+ members
- Awarded silver honor: 2025 International Astronomy & Astrophysics Competition (top 8% of 12,300 people)

**Administrator & Tutor - Metis Project** — September 2024 - June 2026
- Supervised 50 tutors and 200 students for NYC specialized high school entrance exam (SHSAT)
- Analyzed yearly aggregate diagnostic exam data to improve student performance
- Built a framework categorizing questions to isolate specific error distributions and deliver targeted feedback

**Team Player - School Girls Chess Team** — September 2022 - June 2026
- Represented high school across city, state, and national chess tournaments
- Specialized in positional play and evaluated the risk of tactical sacrifices under strict clock constraints
- Awarded First Place Team at the 2023 All-Girls National Chess Championship

## PROJECTS

**Design For America - STEM Coding Lab** — September 2026 - Present
- Designing data collection and analytics solutions for an organization to analyze exam data of 4,000+ students
- Conducting stakeholder research and interviews to define data requirements and performance metrics

**ScottyLabs - ReserveCMU** — September 2026 - Present
- Collaborating to build a platform that improves university room, study space, and facility reservations
- Designing AI-assisted search, filtering systems, reservation workflows, and user feedback collection

**VPython Gravity Assist Simulation** — June 2026
- Recreated the gravity-assist sequences used by Voyagers 1 and 2, as well as a general mode that allows users to experiment with their own planetary and spacecraft configurations
- Implemented Runge-Kutta numerical integration and adaptive time-stepping in N-body gravitational simulator

## SKILLS

**Software:** Jupyter Notebook, Web VPython, VS Code, LaTeX, Google Suite, Figma
**Computer Languages:** Python, Java, HTML/CSS
**Languages:** English, Spanish, Chinese